Project: Worked examples
Title: Shaft: static, fatigue, keys, critical speed
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## 1. Scope and method

A stepped (optionally hollow) shaft on two bearings carrying radial loads in two planes and a torque with a steady (mean) part and an optional alternating part: bending moment, slope and deflection along it, the fatigue and yield safety factors at the sections the user marks, slopes at the bearings, deflection at the loads, the first lateral critical speed and parallel keys.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Shaft bending moment | the shaft as an S2 beam (pin at A, roller at B) in each plane; $M = \sqrt{(M_y^2 + M_z^2)}$ |
| Stress at a section | $\sigma_a = K_f \cdot 32M_a \cdot d/(\pi(d^4 - d_i^4))$, $\tau_m = K_{fs} \cdot 16T_m \cdot d/(\pi(d^4 - d_i^4))$, $\tau_a = K_{fs} \cdot 16T_a \cdot d/(\pi(d^4 - d_i^4))$; $K_f = 1 + q(K_t - 1)$ |
| Endurance limit | $S_e = k_a \cdot k_b \cdot k_c \cdot k_d \cdot k_e \cdot k_f \cdot S_{e1}$, $S_{e1}$ (the rotating-beam limit $S_e'$) $= 0.5 \cdot S_{ut}$ ($S_{ut} \le 1400$ MPa, else 700 MPa); $k_a = a \cdot S_{ut}$^b, $k_b = 1.24d^{-0.107}$ or $1.51d^{-0.157}$, $k_e = 1 - 0.08 \cdot z_R$ |
| Fatigue (DE-Goodman, bending fully reversed, torque $T_m \pm T_a$) | $1/n_f = \sqrt{(\sigma_a^2 + 3\tau_a^2)}/S_e + \sqrt{3} \cdot \tau_m/S_{ut}$; yield $n_y = S_y/\sqrt{((\sigma_a)^2 + 3\tau_{pk}^2)}$, $\tau_{pk}$ from the larger of $T_m + T_a$ and the peak torque $T_{pk}$ when given |
| Critical speed | Rayleigh $\omega = \sqrt{(g \cdot \Sigma m \cdot y/\Sigma m \cdot y^2)}$, y signed (an overhung mass lifts the span, which lowers the quotient); Dunkerley $1/\omega^2 = \Sigma\, m_i \cdot \delta_{ii}$ (lower bound, used for the ratio to the operating speed) |
| Weights | with $w_{masses}$ the carried masses and the shaft's own mass load the shaft as downward forces m·g at the nodes; without it the loads table is the whole load |
| Parallel key, on the peak torque $T_k$ = the larger of $T_m + T_a$ and $T_{pk}$ | $\tau = 2T_k/(d \cdot w \cdot L)$, $\sigma = 4T_k/(d \cdot h \cdot L)$; required length L = $\max(2T_k \cdot n_{target}/(d \cdot w \cdot 0.577 \cdot S_{y,key}), 4T_k \cdot n_{target}/(d \cdot h \cdot S_{y,min}))$, $S_{y,min}$ the lowest of the key, the shaft and (when entered) the hub; a shoulder is checked on both sides and the lower $n_f$ governs |

Basis: Fatigue, shaft stresses, deflection, critical speeds, keys[1](10th ed., Ch. 6 (§6-9, Table 6-2 Marin surface factors), Ch. 7 (§7-4 to §7-7)); Shaft design[2](6th ed., Ch. 12).

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| segments | Shaft segments (left to right) | 5 row(s) | | drawing: GBX-SH-01 rev C | any |
| $x_A$ | Bearing A position | 20 | mm | drawing: GBX-SH-01 rev C | 0 ... 3000 mm |
| $x_B$ | Bearing B position | 280 | mm | drawing: GBX-SH-01 rev C | 1 ... 3000 mm |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| loads | Radial loads and masses (a carried mass adds no weight unless $w_{masses}$ is on: enter its weight as a force in $F_y$) | 1 row(s) | | spec: Gear mesh forces from the gear calculation | any |
| $w_{masses}$ | Add the weight of the carried masses and of the shaft itself as downward vertical loads (off: the loads table is the whole load; the masses then enter the critical speed only) | no | | default | any |
| T | Torque transmitted, steady or mean part $T_m$ | 0.3 | kN·m | spec: Motor rated torque × service factor (E1) | 0 … 50 kN·m |
| $x_{T1}$ | Torque enters at | 150 | mm | drawing: GBX-SH-01 rev C | 0 … 3000 mm |
| $x_{T2}$ | Torque leaves at (blank: the shaft length, so every section beyond $x_{T1}$ carries the torque) | 330 | mm | drawing: GBX-SH-01 rev C | 0 … 3000 mm |
| n | Operating speed | 1450 | rpm | spec: Motor nameplate | 0 … 30000 rpm |
| sections | Sections to check | 4 row(s) | | assumption: Shoulders r/d ≈ 0.05; keyseat at the pinion (K_t 2.14, K_ts 3.0); q from the Neuber fit. | any |
| $S_{ut}$ | Ultimate tensile strength | 700 | MPa | spec: EN 10083-3 42CrMo4 +QT, certificate | 300 … 2000 MPa |
| $S_y$ | Yield strength | 500 | MPa | spec: EN 10083-3 42CrMo4 +QT, certificate | 150 … 1900 MPa |
| E | Young's modulus | 207 | GPa | assumption: Steel. | 50 … 250 GPa |
| ρ | Density (shaft self-mass in the critical speed) | 7850 | kg/m³ | assumption: Steel. | 0 … 20000 kg/m³ |
| finish | Surface finish | machined or cold-drawn | | drawing: GBX-SH-01 rev C | any |
| R | Reliability | 99 | % | assumption: Design reliability. | 50 … 99.9 % |
| $k_d$ | Temperature factor $k_d$ | 1 | | assumption: Below 250 °C. | 0.5 … 1.1 |
| $k_f$ | Miscellaneous factor $k_f$ | 1 | | assumption: No other effects. | 0.3 … 1.5 |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| $n_{target}$ | Target safety factor | 1.5 | | spec: Company design standard | 1.2 … 4 |
| $\theta_{max}$ | Slope limit at the bearings | 0.001 | rad | assumption: Deep-groove ball bearing slope limit. | 0.00001 … 0.1 rad |
| $y_{max}$ | Deflection limit at the loads | 0.13 | mm | assumption: Spur gear deflection limit. | 0.001 … 10 mm |
| $x_{key}$ | Key position | 150 | mm | drawing: GBX-SH-01 rev C | 0 … 3000 mm |
| w | Key width | 14 | mm | drawing: GBX-SH-01 rev C | 1 … 100 mm |
| h | Key height | 9 | mm | drawing: GBX-SH-01 rev C | 1 … 100 mm |
| $L_{key}$ | Key length | 40 | mm | drawing: GBX-SH-01 rev C | 1 … 500 mm |
| $S_{y,key}$ | Key yield strength | 400 | MPa | spec: C45 key steel | 150 … 1500 MPa |

| TABLE ROW | VALUES |
|---|---|
| segments[1] | x_end = 40 mm; d = 35 mm; d_i = 0 mm |
| segments[2] | x_end = 120 mm; d = 40 mm; d_i = 0 mm |
| segments[3] | x_end = 180 mm; d = 45 mm; d_i = 0 mm |
| segments[4] | x_end = 260 mm; d = 40 mm; d_i = 0 mm |
| segments[5] | x_end = 340 mm; d = 35 mm; d_i = 0 mm |
| loads[1] | x = 150 mm; F_y = -2.2 kN; F_z = 6 kN; m = 4 kg |
| sections[1] | x = 120 mm; K_t = 1.7; K_ts = 1.5; q = 0.8; q_s = 0.9 |
| sections[2] | x = 150 mm; K_t = 2.14; K_ts = 3; q = 0.8; q_s = 0.9 |
| sections[3] | x = 180 mm; K_t = 1.7; K_ts = 1.5; q = 0.8; q_s = 0.9 |
| sections[4] | x = 260 mm; K_t = 1.6; K_ts = 1.4; q = 0.8; q_s = 0.9 |

## 3. Assumptions

A1. Linear elastic, two simple bearing supports (pin at A, roller at B); bending fully reversed by rotation; torque a steady part $T_m$ plus an alternating part $T_a$ about it (fully alternating when $T_m = 0$), both acting between the torque-in and torque-out positions. (Shigley 10th ed. §7-4.)
A2. No torsional vibration; only the first lateral mode, by Rayleigh with Dunkerley as a lower bound. (Shigley 10th ed. §7-6.)
A3. $K_t$ and $K_{ts}$ come from user records or published fit equations; no stress-concentration chart is reproduced. (No stress-concentration chart is shipped; the factors are entered or derived.)
A4. At a shoulder both sides are checked, each on its own diameter and bore, and the lower fatigue safety factor governs; $k_c = 1$ because the stresses are combined by von Mises. (Shigley 10th ed. §6-9, §7-4.)
A5. The loads table is the whole load: a carried mass adds no weight to the reactions, the moment or the deflection, and neither does the shaft's own weight, unless $w_{masses}$ is on (which adds m·g at the nodes in the vertical plane). (Stated limitation; $w_{masses}$ adds the weights.)

A6. Loads, sections and steps closer together than 1/200 of the shaft length share one node. (Keeps the beam model well conditioned.)
A7. sections (Sections to check) is taken as 4 row(s). (Shoulders r/d ≈ 0.05; keyseat at the pinion ($K_t$ 2.14, $K_{ts}$ 3.0); q from the Neuber fit.)
A8. E (Young's modulus) is taken as 207 GPa. (Steel.)
A9. ρ (Density (shaft self-mass in the critical speed)) is taken as 7850 kg/m^3. (Steel.)
A10. R (Reliability) is taken as 99 %. (Design reliability.)
A11. $k_d$ (Temperature factor $k_d$) is taken as 1. (Below 250 °C.)
A12. $k_f$ (Miscellaneous factor $k_f$) is taken as 1. (No other effects.)
A13. $\theta_{max}$ (Slope limit at the bearings) is taken as 0.001 rad. (Deep-groove ball bearing slope limit.)
A14. $y_{max}$ (Deflection limit at the loads) is taken as 0.13 mm. (Spur gear deflection limit.)

## 4. Working

[18] $R_A = \sqrt{(R_{A,y}^2 + R_{A,z}^2)}$
$= \sqrt{(1.1^2 + -3^2)}$
$= 3.1953$ kN
Radial reaction at bearing A

[19] $R_B = \sqrt{(R_{B,y}^2 + R_{B,z}^2)}$
$= \sqrt{(1.1^2 + -3^2)}$
$= 3.1953$ kN
Radial reaction at bearing B

[20] $d_A$ = seat diameter at A
$= x = 20$ mm
$= 35$ mm
Shaft diameter at bearing A

[21] $d_B$ = seat diameter at B
$= x = 280$ mm
$= 35$ mm
Shaft diameter at bearing B

[22] $M_{max}$ = max $\sqrt{(M_y^2 + M_z^2)}$ along the shaft
$= 37$ stations
$= 0.41539$ kN·m
Largest bending moment

[23] $\theta_A$ = slope at A (two planes)
$= x = 20$ mm
$= 0.0008961$ rad
Slope at bearing A

[24] $\theta_B$ = slope at B (two planes)
$= x = 280$ mm
$= 0.0008961$ rad
Slope at bearing B

[25] $y_{load}$ = largest deflection at a load
$= 0.071775$ mm at 150
$= 0.071775$ mm
Deflection at the loads

[27] $k_a = a \cdot S_{ut}^b$ [1](10th ed. §6-9, Table 6-2)
$= 4.51 \times 700^{-0.265}$ (machined or cold-drawn)
$= 0.79474$
Surface factor

[28] $k_e = 1 - 0.08 \cdot z_R$ [1](10th ed. §6-9)
$= 1 - 0.08 \times 2.3263$
$= 0.81389$
Reliability factor

[29] $S_{e1} = 0.5 \cdot S_{ut}$ ($S_e$')
= 700 MPa
= 350 MPa
Rotating-beam endurance limit
[30] X1 = $\sigma_a = K_f \cdot 32M/(\pi d^3)$, $\tau_m = K_{fs} \cdot 16T_m/(\pi d^3)$, $\tau_a = K_{fs} \cdot 16T_a/(\pi d^3)$; $1/n_f = \sqrt{(\sigma_a^2 + 3\tau_a^2)}/S_e$ + $\sqrt{3}\tau_m/S_{ut}$ [1](10th ed. §7-4)
= x = 120 mm, d = 40 mm, $k_b$ = 0.83561, $S_e$ = 189.17 MPa, $K_f$ = 1.56, $K_{fs}$ = 1.45, M = 319.53 N·m, N·m, $\sigma_a$ = 79.334 MPa, $\tau_m$ = 0 MPa (shoulder: the side with d = 40 mm governs, of 2)
= 2.3845
Section at 120 mm: fatigue safety factor (DE-Goodman)
[31] X2 = $\sigma_a = K_f \cdot 32M/(\pi d^3)$, $\tau_m = K_{fs} \cdot 16T_m/(\pi d^3)$, $\tau_a = K_{fs} \cdot 16T_a/(\pi d^3)$; $1/n_f = \sqrt{(\sigma_a^2 + 3\tau_a^2)}/S_e$ + $\sqrt{3}\tau_m/S_{ut}$ [1](10th ed. §7-4)
= x = 150 mm, d = 45 mm, $k_b$ = 0.82514, $S_e$ = 186.8 MPa, $K_f$ = 1.912, $K_{fs}$ = 2.8, M = 415.39 N·m, 300 N·m, $\sigma_a$ = 88.778 MPa, $\tau_m$ = 46.947 MPa
= 1.6909
Section at 150 mm: fatigue safety factor (DE-Goodman)
[32] X3 = $\sigma_a = K_f \cdot 32M/(\pi d^3)$, $\tau_m = K_{fs} \cdot 16T_m/(\pi d^3)$, $\tau_a = K_{fs} \cdot 16T_a/(\pi d^3)$; $1/n_f = \sqrt{(\sigma_a^2 + 3\tau_a^2)}/S_e + \sqrt{3}\tau_m/S_{ut}$ [1](10th ed. §7-4)
= x = 180 mm, d = 40 mm, $k_b$ = 0.83561, $S_e$ = 189.17 MPa, $K_f$ = 1.56, $K_{fs}$ = 1.45, M = 319.53 N·m, 300 N·m, $\sigma_a$ = 79.334 MPa, $\tau_m$ = 34.616 MPa (shoulder: the side with d = 40 mm governs, of 2)
= 1.9801
Section at 180 mm: fatigue safety factor (DE-Goodman)
[33] X4 = $\sigma_a = K_f \cdot 32M/(\pi d^3)$, $\tau_m = K_{fs} \cdot 16T_m/(\pi d^3)$, $\tau_a = K_{fs} \cdot 16T_a/(\pi d^3)$; $1/n_f = \sqrt{(\sigma_a^2 + 3\tau_a^2)}/S_e + \sqrt{3}\tau_m/S_{ut}$ [1](10th ed. §7-4)
= x = 260 mm, d = 35 mm, $k_b$ = 0.84763, $S_e$ = 191.9 MPa, $K_f$ = 1.48, $K_{fs}$ = 1.36, M = 63.906 N·m, 300 N·m, $\sigma_a$ = 22.47 MPa, $\tau_m$ = 48.465 MPa (shoulder: the side with d = 35 mm governs, of 2)
= 4.2192
Section at 260 mm: fatigue safety factor (DE-Goodman)
[34] $S_e = k_a \cdot k_b \cdot k_c \cdot k_d \cdot k_e \cdot k_f \cdot S_e$' (governing section)
= $k_c$ = 1
= 186.8 MPa
Endurance limit at the governing section
[35] $n_{f,min}$ = min over sections
= 4 section(s)
= 1.6909
Lowest fatigue safety factor
[36] $n_{y,min}$ = min over sections $S_y/\sigma$'_max
= 4 section(s)
= 4.1532
Lowest yield safety factor
[37] $n_c = \sqrt{(g \cdot \Sigma m \cdot y/\Sigma m \cdot y^2)}$ (Rayleigh, y signed) [1](10th ed. §7-6)
= Σm = 7.2337 kg over 37 nodes
= 38275 rpm
First lateral critical speed (Rayleigh)
[38] $n_{c,D} = 1/\omega^2 = \Sigma\, m_i \cdot \delta_{ii}$ (Dunkerley, lower bound) [1](10th ed. §7-6)
= 37 masses
= 37295 rpm
First critical speed, Dunkerley lower bound
[39] $r_c = n_{c,D}/n$
= 37295 / 1450
= 25.721
Critical-speed ratio (Dunkerley lower bound over operating speed)

[44] $\tau_{key} = 2T/(d \cdot w \cdot L)$ [1](10th ed. §7-7)

$= 2 \times 300\ N \cdot m\ /\ (45 \times 14 \times 40)\ mm^3$

$= 23.81\ MPa$

Key shear stress

[45] $\sigma_{key} = 4T/(d \cdot h \cdot L)$ [1](10th ed. §7-7)

$= 4 \times 300\ N \cdot m\ /\ (45 \times 9 \times 40)\ mm^3$

$= 74.074\ MPa$

Key bearing stress

[47] $L_{key,req} = \max(2T \cdot n_{target}/(d \cdot w \cdot 0.577 S_{y,key}),\ 4T \cdot n_{target}/(d \cdot h \cdot S_{y,min}))$

$= n_{target} = 1.5,\ S_{y,min} = 400\ MPa\ (key\ 400,\ shaft\ 500)$

$= 11.111\ mm$

Key length required

[48] $u_{key} = L_{req}/L$

$= 11.111\ /\ 40$

$= 0.27778$

Key utilisation

[49] $u_f = n_{target}/n_{f,min}$

$= 1.5\ /\ 1.6909$

$= 0.88712$

Fatigue utilisation

[50] $u_y = n_{target}/n_{y,min}$

$= 1.5\ /\ 4.1532$

$= 0.36117$

Yield utilisation

[51] $u_{slope} = \max(\theta_A,\ \theta_B)/\theta_{max}$

$= 0.0008961\ /\ 0.001$

$= 0.8961$

Slope utilisation

[52] $u_{defl} = y_{load}/y_{max}$

$= 0.071775\ /\ 0.13\ mm$

$= 0.55212$

Deflection utilisation

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $R_A$ | 3.1953 kN | Reaction at A |
| $R_B$ | 3.1953 kN | Reaction at B |
| $d_A$ | 35 mm | Seat diameter at A |
| $d_B$ | 35 mm | Seat diameter at B |
| $M_{max}$ | 0.41539 kN·m | Largest bending moment |
| $\theta_A$ | 0.0008961 rad | Slope at A |
| $\theta_B$ | 0.0008961 rad | Slope at B |
| $y_{load}$ | 0.071775 mm | Deflection at the loads |
| $S_e$ | 186.8 MPa | Endurance limit |
| $n_{f,min}$ | 1.6909 | Lowest fatigue safety factor |
| $n_{y,min}$ | 4.1532 | Lowest yield safety factor |
| $n_c$ | 38275 rpm | First critical speed (Rayleigh) |

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $n_{c,D}$ | 37295 rpm | Dunkerley lower bound |
| $r_c$ | 25.721 | Critical-speed ratio |
| $\tau_{key}$ | 23.81 MPa | Key shear |
| $\sigma_{key}$ | 74.074 MPa | Key bearing |
| $L_{key,req}$ | 11.111 mm | Key length required |
| $u_{key}$ | 0.27778 | Key utilisation |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Slope at the bearings against limit | $u_{slope}$ = 0.8961 | $u_{slope} \le 1$ | 0.896 | |
| PASS | Fatigue safety factor against target (DE-Goodman) | $u_f$ = 0.88712 | $u_f \le 1$ | 0.887 | [1](10th ed. §7-4) |
| PASS | Deflection at the loads against limit | $u_{defl}$ = 0.55212 | $u_{defl} \le 1$ | 0.552 | |
| PASS | Yield safety factor against target | $u_y$ = 0.36117 | $u_y \le 1$ | 0.361 | |
| PASS | Key length against required | $u_{key}$ = 0.27778 | $u_{key} \le 1$ | 0.278 | |

## 7. Applicability

WARNING — record: Tabulated values not taken from a reference record: $S_{ut}$, $S_y$, E (Shaft material); $\theta_{max}$, $y_{max}$ (Slope and deflection limits). Enter each as one so its source, edition and licence travel with it.
WARNING — mass-weight: m: the carried mass (4 kg) adds no weight to the reactions, the moment or the deflection, and neither does the shaft's own weight; switch $w_{masses}$ on to add them, or enter the weights as forces in $F_y$.
WARNING — torque-span: $x_{T2}$: the torque is transmitted only between $x_{T1}$ = 150 mm and $x_{T2}$ = 330 mm, so the section(s) at 120 mm carry no torque; set $x_{T2}$ (blank: the shaft length) if the torque is carried there.
WARNING — key-hub: $S_{y,hub}$: no hub strength is entered, so the key bearing is checked against the key and the shaft only; the hub is often the weakest part, enter its yield strength.

## 8. Verification

```
Template mech.shaft@1.3.0 Content hash e8975637e45b3669 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

## 9. References

1. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.
2. Mott, Machine Elements in Mechanical Design.